

# THE ZERO PARADOX

## ZP-L: Incomputability Convergence

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All theorems §I–§VII proved sorry-free in Lean 4. Axiom footprint: [propxt, Classical.choice, Quot.sound] throughout this document’s own Lean file (Gentzen.lean). Across the ZP layers the footprints are not uniform — ZP-K Section IV tabulates them.

ZP-L establishes four results connecting the formal axioms of the ZP framework to standard results in ordinal theory and computability. First, the axiom footprints of the layers tabulated in Section I are surveyed, and they are not uniform; ZP-K Section IV tabulates the measured footprints. Second, Rogers’ fixed-point theorem for a total computable transformation (Mathlib fixed\_point; inter-derivable with Kleene’s second recursion theorem, fixed\_point<sub>2</sub>) is formalized as a wrapper, formalizing the computational fixed-point structure. Third, the ordinal  $\varepsilon_0$  is fully characterized as the first fixed point of  $\alpha \mapsto \omega^\alpha$  and the limit of the tower  $\omega, \omega^\omega, \omega^{\omega^\omega}, \dots$ . Fourth, ordinals below  $\varepsilon_0$  encode into  $\mathbb{Z}_2$  via their Cantor normal form, and as the tower stages approach  $\varepsilon_0$ , their 2-adic encodings converge to  $0 = \perp$ .

The central result (§VII) is the canonical snap map:  $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$  simultaneously satisfies all five conditions — monotone, tower-aligned, fixed-point-respecting, snapping at  $\varepsilon_0$ , and  $\varepsilon_0$  minimal. All conditions are verified without free hypotheses for this witness. Proved sorry-free, axiom footprint [propxt, Classical.choice, Quot.sound] throughout ZeroParadox/Ordinal/Gentzen.lean, measured.

### Section I: Axiom Footprint Convergence

Non-constructibility appears across the layers listed below. The axiom footprints are not uniform: ZP-K Section IV tabulates the measured footprints, and the ZPJ/K row below names a witness on each side — bot\_self\_mem, which measures no axioms, and botCode, which carries them.

| Layer | Formal Language          | Expression of non-constructibility                      |
|-------|--------------------------|---------------------------------------------------------|
| ZPB   | Topology                 | C3: no continuous path $\perp \rightarrow x \neq \perp$ |
| ZPC   | Information Theory       | L-INF: infinite surprisal at $\perp$                    |
| ZPJ/K | Set Theory + Computation | bot_self_mem (AFA); botCode (Kleene)                    |
| ZPI   | Algorithmic IT           | $K(S_n n)/ S_n  \rightarrow 1$ ; K uncomputable         |

#### Remark: K is not computed in Lean

Kolmogorov complexity K is not computed in Lean in this framework. The AFA/Kleene route reaches the same fixed-point structure via a path whose Kleene step is a KleeneStructure requirement, without requiring K to be explicitly computed.

### Remark: K is not computed in Lean

Axiom footprint evidence — the following ZP-K theorems all carry [propext, Classical.choice, Quot.sound]:

t\_comp (T-COMP three-way equivalence; the computational face is an assumption, not a clause)

da1\_paths\_unified

isComputationalQuine\_undecidable

infinite\_quine\_family

Gentzen.lean carries that footprint throughout, measured — and it is not inherited from the theorems above. Mathlib's ordinal theory carries it on its own: Ordinal.nfp and Ordinal.epsilon\_zero\_eq\_nfp each measure [propext, Classical.choice, Quot.sound] with no computability module in scope. ZP-K Section IV tabulates the measured footprints for the computability side.

## Section II: Rogers Fixed-Point Stability

Rogers' fixed-point theorem (Mathlib fixed\_point; inter-derivable with Kleene's second recursion theorem, fixed\_point<sub>2</sub>) states that any total computable transformation of a code has a behavioral fixed point: a code  $c$  such that running  $f(c)$  and running  $c$  produce the same partial function. For any computable transformation, at least one fixed-point code exists.

### Theorem: roger\_fixed\_point\_stability (Gentzen.lean §II)

For any computable  $f : \text{Code} \rightarrow \text{Code}$ ,

$\exists c : \text{Code}, \text{eval } (f\ c) = \text{eval } c$

Proof: wrapper around roger\_fixed\_point\_exists.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Note: this is an existential result. The specific code  $c$  is not constructively produced. ZP-K Section IV tabulates the measured footprints.

## Section III: The Ordinal $\epsilon_0$

The ordinal  $\epsilon_0$  is the smallest fixed point of the map  $\alpha \mapsto \omega^\alpha$ . It is the supremum of the tower  $0, 1, \omega, \omega^\omega, \omega^{\omega^\omega}, \dots$  — each stage is strictly below  $\epsilon_0$ , and  $\epsilon_0$  is not reached by any finite iteration. This section is entirely within Lean scope via Mathlib's ordinal machinery.

### Definitions (Gentzen.lean §III)

epsilonZero : Ordinal := Ordinal.epsilon 0 (= nfp ( $\omega^\cdot$ ) 0 = veblen 1 0)

fundamentalSeq :  $\mathbb{N} \rightarrow \text{Ordinal} := \text{fun } n \mapsto (\alpha \mapsto \omega^\alpha)^{[n]} 0$

## Definitions (Gentzen.lean §III)

Explicit stages:

$$\text{fundamentalSeq } 0 = 0$$

$$\text{fundamentalSeq } 1 = 1$$

$$\text{fundamentalSeq } 2 = \omega$$

$$\text{fundamentalSeq } 3 = \omega^\omega$$

$$\text{fundamentalSeq } (n+1) = \omega^{(\text{fundamentalSeq } n)}$$

## Theorem: epsilonZero\_fixedPoint

$$\omega^{\varepsilon_0} = \varepsilon_0$$

Proof: Ordinal.omega0\_opow\_epsilon 0 (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

## Theorem: epsilonZero\_eq\_nfp

$$\varepsilon_0 = \text{nfp } (\omega^{\cdot}) 0$$

Proof: Ordinal.epsilon\_zero\_eq\_nfp (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

## Theorem: epsilonZero\_eq\_iSup

$$\varepsilon_0 = \sqcup n : \mathbb{N}, \text{fundamentalSeq } n$$

Proof: iSup\_iterate\_eq\_nfp (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

## Theorem: epsilonZero\_tower\_lt

$$\forall n : \mathbb{N}, \text{fundamentalSeq } n < \varepsilon_0$$

Every finite stage of the tower is strictly below  $\varepsilon_0$ .

Proof: Ordinal.iterate\_omega0\_opow\_lt\_epsilon\_zero (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

## Theorem: epsilonZero\_le\_fixedPoint

$$\forall b : \text{Ordinal}, \omega^b = b \rightarrow \varepsilon_0 \leq b$$

### Theorem: epsilonZero\_le\_fixedPoint

$\epsilon_0$  is the least fixed point of  $\omega^\cdot$  above 0.

Proof: Ordinal.epsilon\_zero\_le\_of\_omega0\_opow\_le (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: fundamentalSeq\_strictMono

$\forall n : \mathbb{N}, \text{fundamentalSeq } n < \text{fundamentalSeq } (n + 1)$

The tower is strictly monotone.

Proof: by induction; successor step uses isNormal\_opow.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Remark R-L.1: Proof-Theoretic Alignment

Gentzen's theorem establishes that  $\epsilon_0$  is the proof-theoretic ordinal of Peano Arithmetic — the furthest that PA's proofs of transfinite induction reach — and it takes more than one result. Transfinite induction up to  $\epsilon_0$  suffices to prove PA consistent (1936). That induction is not itself provable in PA, which follows indirectly from Gödel's theorem together with the 1936 result, and which Gentzen proved directly in 1943. And Gentzen reports, crediting Hilbert-Bernays, that transfinite induction up to any ordinal strictly below  $\epsilon_0$  is provable in PA. Sufficiency scales upward, so the 1936 bound alone holds of any larger ordinal. What pins the value to exactly  $\epsilon_0$  is the other two together, and they run in opposite directions: PA proves transfinite induction at every ordinal strictly below  $\epsilon_0$ , and does not prove it at  $\epsilon_0$  itself. Neither half alone locates the boundary. This is not claimed or proved here.

ZP-L derives  $\epsilon_0$  as the snap threshold from ordinal fixed-point structure, independently of proof theory. Both derivations locate the same boundary: the ordinal where  $\omega$ -tower self-iteration becomes self-limiting. No claim is made that this alignment is more than a structural observation.

### Remark R-L.2: Surreal Numbers

Every ordinal is a surreal number (Conway, 1976), so  $\epsilon_0$  lives in the surreal number field No. No satisfies the axioms of a real-closed field, and by Tarski's completeness theorem for the theory of real-closed fields, every first-order sentence in the language of ordered fields that holds in  $\mathbb{R}$  also holds in No, and vice versa.

ZP-F (The Counterexamples) proves that the Binary Snap — the transition from  $\perp$  to the first non-null ordinal threshold ( $\epsilon_0$ , established in §V–§VII above) — cannot occur in any linearly ordered field. This result applies directly to No considered as a linearly ordered field. Note the scope: what is derived is the transition's SHAPE and its impossibility in an ordered field.

The surreals therefore contain  $\epsilon_0$  as an ordinal while simultaneously satisfying the density condition that blocks the Binary Snap in their ordered field structure. Both structures coexist in No:  $\epsilon_0$  is present as an ordinal, and the field density that blocks the snap is present in the field structure. The two results apply to different structural aspects of No.

### Remark R-L.3: Hyperreals and Łoś's Theorem

The hyperreals  ${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/U$  (ultrafilter  $U$  on  $\mathbb{N}$ ) satisfy a transfer result: by Łoś's theorem, every first-order sentence true in  $\mathbb{R}$  holds in  ${}^*\mathbb{R}$ . Łoś's theorem applies to the full first-order theory, so any first-order property of  $\mathbb{R}$  — including field density — transfers.

Field density transfers: between any two hyperreals there is another. The density condition that blocks the Binary Snap in  $\mathbb{R}$  (proved in ZP-F) therefore holds in  ${}^*\mathbb{R}$  as well.

The non-standard naturals  ${}^*\mathbb{N}$  contain infinite elements, but every infinite  $H \in {}^*\mathbb{N}$  has a predecessor  $H-1$  in  ${}^*\mathbb{N}$ .  $\epsilon_0$  is a limit ordinal: it has no predecessor and is the supremum of the tower below it.  ${}^*\mathbb{N}$  cannot host the Binary Snap not only because of field density, but because its infinite elements are successor-like rather than limit-like. The snap requires genuine limit ordinal structure.

The monad of 0 in  ${}^*\mathbb{R}$  — the external set  $\{\epsilon : |\epsilon| < 1/n \text{ for all standard } n\}$  — is external to the ultrapower. Between any two distinct elements of the monad there is another (their arithmetic mean in  ${}^*\mathbb{R}$ ), so no discrete jump at 0 is possible in  ${}^*\mathbb{R}$ .

Field density is an internal property of the ordered field structure of  ${}^*\mathbb{R}$  and transfers by Łoś's theorem. Limit ordinal structure is set-theoretic and is not preserved by the ultrapower construction:  ${}^*\mathbb{N}$  contains only successor-like infinite elements, not limit-like ones. The two results — transfer of density, failure of limit-ordinal structure — come from different theorems.

## Section IV: Cantor Normal Form Bridge

Every ordinal below  $\epsilon_0$  has a unique Cantor normal form — a finite sum  $a_1 \cdot \omega^{e_1} + a_2 \cdot \omega^{e_2} + \dots$  with strictly decreasing exponents  $e_1 > e_2 > \dots$  and each coefficient a nonzero natural number (a nonzero ordinal strictly below  $\omega$ ). In Lean: `NONote (Mathlib.SetTheory.Ordinal.Notation)`. The encoding `cnfToZp2` maps each such ordinal to  $\mathbb{Z}_2$  via structural recursion on the Cantor normal form.

### Definition: `cnfToZp2` (Gentzen.lean §IV)

`cnfToZp2 : NONote →  $\mathbb{Z}_2$`

Base: `cnfToZp2(0) = 0`

Recursive: `cnfToZp2( $\omega^e \cdot n + a$ ) =  $2^{(v_2(\text{cnfToZp2}(e)) + 1)} \cdot n + \text{cnfToZp2}(a)$  [ $n : \mathbb{N}^+$ ]`

where  $v_2$  denotes the 2-adic valuation.

Valuation of the  $n$ -th tower stage:

`cnfToZp2(towerNONote 0) = 0` (Lean: `PadicInt.valuation 0 = 0`; standard:  $v_2(0) = +\infty$ )

`cnfToZp2(towerNONote 1) = 2` (valuation 1)

`cnfToZp2(towerNONote 2) = 4` (valuation 2)

`cnfToZp2(towerNONote n) =  $2^n$`  (valuation  $n$ )

| Theorem: <code>cnfToZp2_tower_valuation</code>                                            |
|-------------------------------------------------------------------------------------------|
| $\forall n : \mathbb{N}, (\text{cnfToZp2 } (\text{towerNONote } n)).\text{valuation} = n$ |
| The 2-adic valuation of the $n$ -th tower stage encoding equals $n$ .                     |
| Proof: by induction; successor step uses <code>PadicInt.valuation_pow</code> .            |
| Lean purity: <code>[propext, Classical.choice, Quot.sound]</code> . ✓                     |

| Theorem: <code>cnfToZp2_valuation_unbounded</code>                                                          |
|-------------------------------------------------------------------------------------------------------------|
| $\forall k : \mathbb{N}, \exists \alpha : \text{NONote}, k \leq (\text{cnfToZp2 } \alpha).\text{valuation}$ |
| The 2-adic valuation is unbounded across ordinals below $\varepsilon_0$ .                                   |
| Proof: <code>towerNONote k</code> witnesses the bound.                                                      |
| Lean purity: <code>[propext, Classical.choice, Quot.sound]</code> . ✓                                       |

| Theorem: <code>tower_converges_to_zero</code>                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>Filter.Tendsto (fun n ↦ cnfToZp2 (towerNONote n)) Filter.atTop (nhds 0)</code>                                                                                                                                                                        |
| The tower encodings converge to $0 = \perp$ in $\mathbb{Z}_2$ .                                                                                                                                                                                             |
| Proof: each stage <code>cnfToZp2(towerNONote (n+1)) = 2<sup>^(n+1)</sup></code> in $\mathbb{Z}_2$ , so its 2-adic norm is $\ 2\ ^{(n+1)} = (1/2)^{(n+1)} \rightarrow 0$ . Uses <code>Metric.tendsto_atTop</code> and <code>exists_pow_lt_of_lt_one</code> . |
| Lean purity: <code>[propext, Classical.choice, Quot.sound]</code> . ✓                                                                                                                                                                                       |

## Section V: Ordinal Tower Limit and Snap Threshold

This section connects the ordinal tower to the ZPE machine-phase snap. The cofinality theorem shows that the fundamental sequence approaches  $\varepsilon_0$  from below. The lower-bound theorem shows that any monotone tower-aligned map must assign  $c_0$  to all pre- $\varepsilon_0$  ordinals. A canonical witness snapping exactly at  $\varepsilon_0$  is exhibited.

| What this does NOT claim (§V)                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gentzen’s theorem: that $\varepsilon_0$ is the proof-theoretic ordinal of PA (not claimed — see Remark R-L.1).                                                                                                                                                                        |
| Any statement about formal provability in PA.                                                                                                                                                                                                                                         |
| That $\varepsilon_0$ is the UNIQUE minimal snap boundary: <code>snap_threshold_is_epsilon_zero</code> shows no ordinal below $\varepsilon_0$ works (for maps satisfying the stated hypotheses), but does not rule out maps that snap at some ordinal strictly above $\varepsilon_0$ . |
| That the snap threshold result applies to all maps <code>Ordinal → MachinePhase</code> regardless of the monotonicity and tower-alignment hypotheses.                                                                                                                                 |

### Theorem: fundamentalSeq\_cofinal

$\forall \alpha : \text{Ordinal}, \alpha < \varepsilon_0 \rightarrow \exists n : \mathbb{N}, \alpha < \text{fundamentalSeq } n$

The fundamental sequence is cofinal in  $\varepsilon_0$ : for any ordinal below  $\varepsilon_0$ , some tower stage exceeds it.

Proof:  $\varepsilon_0 = \text{nfp } (\omega^\cdot) 0$  (`epsilonZero_eq_nfp`), and `lt_nfp_iff` gives  $a < \text{nfp } f \ b \leftrightarrow \exists n, a < f^n[n] \ b$ .

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

### Theorem: snap\_threshold\_is\_epsilon\_zero

For any  $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$  satisfying:

(a) `hmono`:  $\forall \alpha \leq \beta, \text{join } (\phi \ \alpha) (\phi \ \beta) = \phi \ \beta$  (order-non-decreasing in the `ZPSemilattice` sense:  $\phi \ \alpha \leq \phi \ \beta$ )

(b) `h0`:  $\forall n : \mathbb{N}, \phi \ (\text{fundamentalSeq } n) = c_0$

we have:  $\forall \alpha < \varepsilon_0, \phi \ \alpha = c_0$

This is a lower bound: no ordinal below  $\varepsilon_0$  is a snap point for maps satisfying (a) and (b). "Minimal" not "unique."

Proof: cofinality gives a stage above  $\alpha$ ; monotonicity chains  $\phi \ \alpha \leq \phi(\text{stage}) = c_0$ ;  $c_0 = \perp$  forces  $\phi \ \alpha = c_0$ .

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

### Theorem: c1\_epsilon\_zero\_identification

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall n : \mathbb{N}, \phi \ (\text{fundamentalSeq } n) = c_0) \wedge \phi \ \varepsilon_0 = c_1$

Witness:  $\phi \ \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$ .

One specific map sends all tower stages to  $c_0$  and  $\varepsilon_0$  to  $c_1$ .

The stronger structural claim — an order-preserving morphism  $\text{Ordinal} \rightarrow_o \text{MachinePhase}$  compatible with the  $\text{CNF} \rightarrow \mathbb{Z}_2$  encoding — remains outside Lean scope: no type bridge between `Ordinal` and `MachinePhase` is defined in this library.

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

## Section VI: Kleene-Ordinal Fixed-Point Bridge

The ordinal fixed-point structure ( $\varepsilon_0 = \text{nfp } (\omega^\cdot) 0$ ,  $\omega^\cdot \varepsilon_0 = \varepsilon_0$ ) and the computational fixed-point structure (Kleene's recursion theorem, `roger_fixed_point_stability`) both carry `Classical.choice` in the proofs recorded here — parallel structure, not a proved isomorphism. The hypothesis `hfp` encodes that ordinal fixed points of  $\omega^\cdot$  map to the snap state  $c_1$ . Under this hypothesis plus monotonicity and tower alignment,  $\varepsilon_0$  is the minimal snap threshold.

### Theorem: snap\_exactly\_at\_epsilon\_zero

For any  $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$  satisfying:

(a) hmono: order-non-decreasing ( $\text{join } (\phi \alpha) (\phi \beta) = \phi \beta$  for  $\alpha \leq \beta$ )

(b) h0: every tower stage maps to  $c_0$

(c) hfp: every fixed point of  $\omega^\cdot$  maps to  $c_1$

we have:  $\phi \varepsilon_0 = c_1$  AND  $\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$

$\varepsilon_0$  is the minimal snap threshold:  $\phi$  assigns  $c_1$  first at  $\varepsilon_0$ .

Note: "minimal" not "unique." A map satisfying these hypotheses could also assign  $c_1$  to ordinals above  $\varepsilon_0$ ; what is ruled out is any snap strictly before  $\varepsilon_0$ .

Proof: upper bound from hfp + epsilonZero\_fixedPoint; lower bound from snap\_threshold\_is\_epsilon\_zero.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: kleene\_ordinal\_snap\_bridge

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0) \wedge$

$(\forall \alpha, \omega^\alpha \alpha = \alpha \rightarrow \phi \alpha = c_1) \wedge$

$\phi \varepsilon_0 = c_1 \wedge$

$\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$

Witness:  $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$ .

Note: the theorem name refers to the informal §VI conceptual parallel between Kleene recursion fixed points and ordinal fixed points of  $\omega^\cdot$ . The theorem itself is purely ordinal — no Code or eval appears.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

## Section VII: Canonical Snap Map — Full Closure

The canonical threshold map  $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$  satisfies all three conditions of snap\_exactly\_at\_epsilon\_zero (hmono, h0, hfp). For this specific map, the snap identification is unconditional: all five conditions are simultaneously verified without free hypotheses.

### Theorem: snap\_map\_mono

$\forall \alpha \beta : \text{Ordinal}, \alpha \leq \beta \rightarrow$

$\text{join } (\text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1)$

### Theorem: snap\_map\_mono

(if  $\beta < \varepsilon_0$  then  $c_0$  else  $c_1$ ) =

if  $\beta < \varepsilon_0$  then  $c_0$  else  $c_1$

The canonical map is order-non-decreasing.

Proof: four cases ( $\alpha/\beta$  vs  $\varepsilon_0$ ). The case  $\alpha \geq \varepsilon_0$  with  $\beta < \varepsilon_0$  is impossible by  $\alpha \leq \beta$ . Each live case closes by rfl from the MachinePhase join definition.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: epsilon\_zero\_snap\_canonical

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall \alpha \beta, \alpha \leq \beta \rightarrow \text{join}(\phi \alpha)(\phi \beta) = \phi \beta) \wedge [\text{hmono}]$

$(\forall n : \mathbb{N}, \phi(\text{fundamentalSeq } n) = c_0) \wedge [\text{h0}]$

$(\forall \alpha, \omega^\alpha = \alpha \rightarrow \phi \alpha = c_1) \wedge [\text{hfp}]$

$\phi \varepsilon_0 = c_1 \wedge [\text{snap}]$

$\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$  [minimality]

All five conditions verified for the explicit witness  $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$ , with no free hypotheses.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

### Theorem: snap\_zp2\_correspondence

Four independent facts about the same tower sequence:

(i)  $\forall n : \mathbb{N}, \text{fundamentalSeq } n < \varepsilon_0$

(ii)  $\forall n : \mathbb{N}, \phi(\text{fundamentalSeq } n) = c_0$  where  $\phi$  is the canonical map

(iii)  $\text{Filter.Tendsto}(\text{fun } n \mapsto \text{cnfToZp2}(\text{towerNONote } n)) \text{ Filter.atTop (nhds } 0)$

(iv)  $\phi \varepsilon_0 = c_1$

The same tower sequence witnesses both the ordinal approach to  $\varepsilon_0$  and the 2-adic approach to 0. The full structural identification ( $\varepsilon_0 \leftrightarrow \perp$  via a type bridge) remains outside Lean scope — see §V.

Proof:  $\langle \text{epsilonZero\_tower\_lt}, \text{fun } n \mapsto \text{if\_pos } \dots, \text{tower\_converges\_to\_zero}, \text{if\_neg } (\text{lt\_irrefl } \varepsilon_0) \rangle$

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

## Remaining Formal Gap

## The Remaining Gap

The identification of ZPE's MachinePhase element  $c_1$  with the ordinal  $\epsilon_0$  via a formal type bridge remains outside Lean scope.

What is proved: a canonical map  $\text{Ordinal} \rightarrow \text{MachinePhase}$  assigns  $c_1$  exactly at  $\epsilon_0$  and nowhere earlier. What is not proved: a canonical ZPSemilattice morphism  $\text{MachinePhase} \rightarrow \mathbb{Z}_2$  that would connect ZPE's  $\perp = c_0$  to ZPB's  $\perp = 0$  formally.

Two of the three pieces such a bridge was expected to need are built, in ZPM (ZeroParadox/Ordinal/Incompleteness.lean §I–§II), and the third is the one still open:

(1)  $\text{snapEmbed} : \text{MachinePhase} \rightarrow \mathbb{Z}_2$  exists, sending  $c_0 \mapsto 1$  and  $c_1 \mapsto 0$  — the snap state to 2-adic zero, the pre-snap state to a unit ( $\text{snapEmbed\_c0}$ ,  $\text{snapEmbed\_c1}$ , both by rfl)

(2)  $\text{snapEmbed\_mul\_morphism}$  proves  $\text{snapEmbed}(\text{join } a \text{ } b) = \text{snapEmbed } a \times \text{snapEmbed } b$ , carrying join to multiplication because  $c_1$  absorbs joins as 0 absorbs products. That is an absorbing-element morphism, not a ZPSemilattice morphism:  $\mathbb{Z}_2$  carries no lattice  $\perp$

(3) still open — no theorem derives the alignment from  $\text{tower\_converges\_to\_zero}$  through  $\text{snapEmbed}$

So  $\text{hfp}$  is derived rather than assumed:  $\text{hfp\_from\_epsilon\_zero}$  obtains it from monotonicity together with  $\text{h}\epsilon_0 : \phi \epsilon_0 = c_1$ , and  $\text{snap\_unconditional}$  uses it in place of the free hypothesis of  $\text{snap\_exactly\_at\_epsilon\_zero}$ . What remains open is  $\text{h}\epsilon_0$  itself — deriving  $\phi \epsilon_0 = c_1$  from the 2-adic structure rather than taking it as an alignment hypothesis, which ZPM §II names the Classical.choice inversion conjecture. The canonical witness ( $\text{epsilon\_zero\_snap\_canonical}$ ) satisfies all five conditions without any such bridge; what the bridge would close is the gap between the two formal instances of  $\perp$  across ZPE and ZPB.

## Theorem Summary

| Theorem                        | Section | Status   |
|--------------------------------|---------|----------|
| roger_fixed_point_stability    | §II     | Proved ✓ |
| epsilonZero_fixedPoint         | §III    | Proved ✓ |
| epsilonZero_eq_nfp             | §III    | Proved ✓ |
| epsilonZero_eq_iSup            | §III    | Proved ✓ |
| epsilonZero_tower_lt           | §III    | Proved ✓ |
| epsilonZero_le_fixedPoint      | §III    | Proved ✓ |
| fundamentalSeq_strictMono      | §III    | Proved ✓ |
| tower_stage_zero / _one / _two | §III    | Proved ✓ |
| towerNONote_repr               | §IV     | Proved ✓ |
| cnfToZp2_tower_valuation       | §IV     | Proved ✓ |

| Theorem                         | Section | Status   |
|---------------------------------|---------|----------|
| cnfToZp2_valuation_unbounded    | §IV     | Proved ✓ |
| fundamentalSeq_zp2_converges    | §IV     | Proved ✓ |
| tower_converges_to_zero         | §IV     | Proved ✓ |
| zpe_snap_ordinal_correspondence | §V      | Proved ✓ |
| epsilonZero_tower_bound         | §V      | Proved ✓ |
| c1_epsilon_zero_identification  | §V      | Proved ✓ |
| fundamentalSeq_cofinal          | §V      | Proved ✓ |
| snap_threshold_is_epsilon_zero  | §V      | Proved ✓ |
| snap_exactly_at_epsilon_zero    | §VI     | Proved ✓ |
| kleene_ordinal_snap_bridge      | §VI     | Proved ✓ |
| snap_map_mono                   | §VII    | Proved ✓ |
| epsilon_zero_snap_canonical     | §VII    | Proved ✓ |
| snap_zp2_correspondence         | §VII    | Proved ✓ |

## Axiom Purity

Every theorem in the Theorem Summary above carries axiom footprint: [propext, Classical.choice, Quot.sound]. Section I's table and ZP-K Section IV record where the ZP layers differ — Section I's own ZPJ/K row names bot\_self\_mem, which measures no axioms.

These are standard Mathlib infrastructure axioms (ordinal theory, p-adic analysis, computability). They are not ZP-L commitments.

The footprints are not uniform; ZP-K Section IV tabulates the measurements. In the computability layer Classical.choice is reached through Mathlib's numbering of program codes (Denumerable Code): it is present even in a computable instance with a constant code. ZP-K's machinePhaseKleene also picks botCode with Classical.choose, which is what makes that instance noncomputable. Whether the numbering's footprint is essential is not measured. Its presence is expected and documented.

Zero sorry in Gentzen.lean. Verified: lake build, May 2026.